

EDUCATION AND SCHOLASTIC ACHIEVEMENTS				
Program		Institute	CGPA	Year
B.Tech (Honors), Electrical Engineering		Indian Institute of Technology, Madras	9.30/10	2023-27
Class XII, Class X (CBSE)		National Public School HSR, Bangalore	96.6%, 96.8%	2023, 2021
- Recipient of the MV Undergraduate Prize awarded to the female student with the best all-round performance up to 6 semesters. - AIR 132 in JEE Mains 2023 out of 11.6 lakh candidates; ranked 3rd nationally among girls. AIR 1632 in JEE Advanced 2023. - AIR 14 in UG Common Entrance Exam for Design 2023 among 14,000 candidates. SAT score of 1580/1600 in 2023 (99th %ile).				
RELEVANT COURSEWORK				
Digital Signal Processing (#)	Foundations of ML	Information Theory (#)	Microprocessors (#)	Foundations of Probability
Communication Systems	Adv. Topics in AI (*)	Estimation Theory (*)	Computer Organization	DSA in Python (#)
RF & Optical Communications	Comm. Networks (#)	Convex Optimization (*)	Quantum Comp. & Info.	Numerical Methods in C
SKILLS				
Languages	Python, C++, C, MATLAB, AVR Assembly, ARM Assembly, Verilog, Dart, HTML, CSS, Bash, JavaScript, SQL			
ML Techniques and Libraries	Deep Learning, CNNs, RNNs, LSTMs, Transformers, Attention, Autoencoders, Time-Series Prediction, Reinforcement learning, Hyperparameter Optimization, Pytorch, TensorFlow, Keras, Scikit-learn, Optuna, NumPy, SciPy, Pandas			
Tools	Wireshark, Code Composer Studio, Git, Xilinx Vivado, LT Spice, Keil Microvision, Microchip Studio, ARM Cortex-M			
PUBLICATIONS				
Optics Express Journal, Volume 34, First Author Paper (June 2026) (Link)	Attention-assisted deep neural network for nonlinear equalization & denoising in time-stretched photonic ADCs			
	- Developed a signal recovery methodology in photonic ADCs for distortion mitigation using a custom deep neural architecture combining attention-enhanced LSTM and ANNs, reducing ADC-bandwidth requirement by 3x .			
	- Presented at EE Research Scholars' Symposium at IIT Madras after being selected as a top UG research project .			
RESEARCH PROJECTS AND COMPETITIONS				
Adaptive Equalization: Optimizing DSP Algorithms (Link)	Undergraduate Research January - May 2025; Guides: Prof. LN Theagarajan, Prof. D Venkitesh			
	- Implemented modifications in adaptive equalization algos. for coherent optical comm. systems achieving 8x faster convergence. Developed algo. with 95% accuracy for simultaneous compensation of dispersion and freq. offset.			
RL-based Frameworks for Fiber Optical Impairment Correction (Link)	Undergraduate Research July - December 2025; Guides: Prof. LN Theagarajan, Prof. D Venkitesh			
	- Developed a single compensation system using RL frameworks to correct receiver impairments in optical channels. Formulated MAB framework for polarization & dispersion compensation in 2x2 MIMO fibre-optic systems.			
Adaptive Channel Sensing & Client- Centric Radio Resource Management (Link)	- Built UCB-based filter tap selection ; converged to optimal configuration from 5L+ options in ~1000 rounds.			
	Extended to Q-Learning using experience replay & teacher training from classical CMA; development in progress.			
Communication System Simulations (EE3005; 2026) (Link)	Inter-IIT Tech Meet 2025 conducted by IIT Patna: Problem Statement posed by Arista Networks			
	- Developed an MAB algorithm for adaptive channel scheduling based on channel noise, replacing naive periodic scanning. Achieved 3.99% improvement in channel power estimation when evaluated on correlated I/Q data.			
Wi-fi Coverage Review (EE5150; 2026) (Link)	- Built an end-to-end additional-radio sensing pipeline combining CNN-based interference classification, MAB-based adaptive scanning, and CUSUM change detection for automated channel-switch recommendations .			
	- Simulated communication systems including QAM, DSB-SC AM, and FM modulation/demodulation in Python.			
Teaching Asst. (2025)	- Implemented Delay-and-Sum beamforming algorithm in Python for spatial signal reconstruction from microphone array data, applying inter-element time-delay compensation for source localization in antenna-array systems.			
	- Conducted WLAN site surveys across 32 locations, analysing ~9,600 beacon frames from PCAP traces to fit an empirical indoor path-loss model. Estimated propagation parameters (RSSI, path-loss exponent, attenuation).			
PROFESSIONAL EXPERIENCE				
Texas Instruments Summer Intern: Embedded SW (2026)	- Summer intern in the Automotive Applications Software team for the F29x microcontroller to satisfy marketing requirements. Worked on protecting flash memory and loading multi-core programs in single-core flash.			
	- Demonstrated training CNNs for power grid fault detection in TI EdgeAI Studio under fixed latency & memory.			
Texas Instruments WISH Scholar (2025)	- Selected as one of 40 among 2000+ applicants nationwide for month-long mentorship in the embedded software track ; Worked on programming PWM-controlled stepper motor systems with sensors and UART on C2000 MCUs.			
Teaching Asst. (2025)	- TA for Signals & Systems for 160+ students. Conducted hands-on sessions to program an audio equalizer .			
POSITIONS OF RESPONSIBILITY				
Product Design Club Head (2025)	- Led 50+ members, oversaw 9 interdisciplinary projects; Collaborated with IISc, JIPMER for projects and IITK, IITKGP, IITD for events. Secured 2 Patents under the club; led Freshie Design Challenge with 300+ participants .			
'InstiLite' Founder Product Design Club	- Founded and developed InstiLite, a full-stack mobile platform for event discovery, scheduling, and application workflows; implemented backend services with NodeJS and SQL architecture ; used Flutter for client interfaces.			
Placement Cell (2025)	- As Coordinator, planned executive framework for placement drive for 150+ students; onboarded 3 companies.			
EXTRA-CURRICULARS				
Technovation Ambassador (2021)	- India-wide ambassador for Technovation, a California-based program for school girls, addressing gender gaps in tech & entrepreneurship. Mentored 200+ girls over a year . Conducted sessions in design thinking & programming.			
Times Of India Summer Internship	- Selected as school representative to be one among 40 students chosen for a 2-week journalism internship.			
	- Selected as guest editor by Times of India for the Newspaper In Education Guest Edition in 2020.			
Abbreviations- UG: Undergraduate, SW: Software, ADCs: Analog-to-digital converters, MAB: Multi-armed bandits, LSTM: Long short-term neural network, CNN: Convolutional Neural Network, RNNs: Recurrent Neural Network, ANNs: Artificial Neural Network. (#): Received highest grade; (*): Ongoing course;				